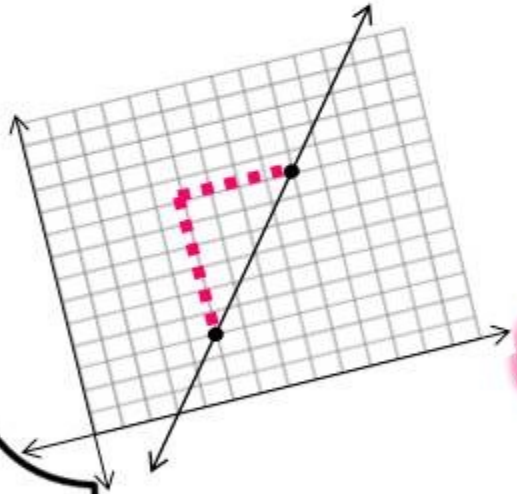
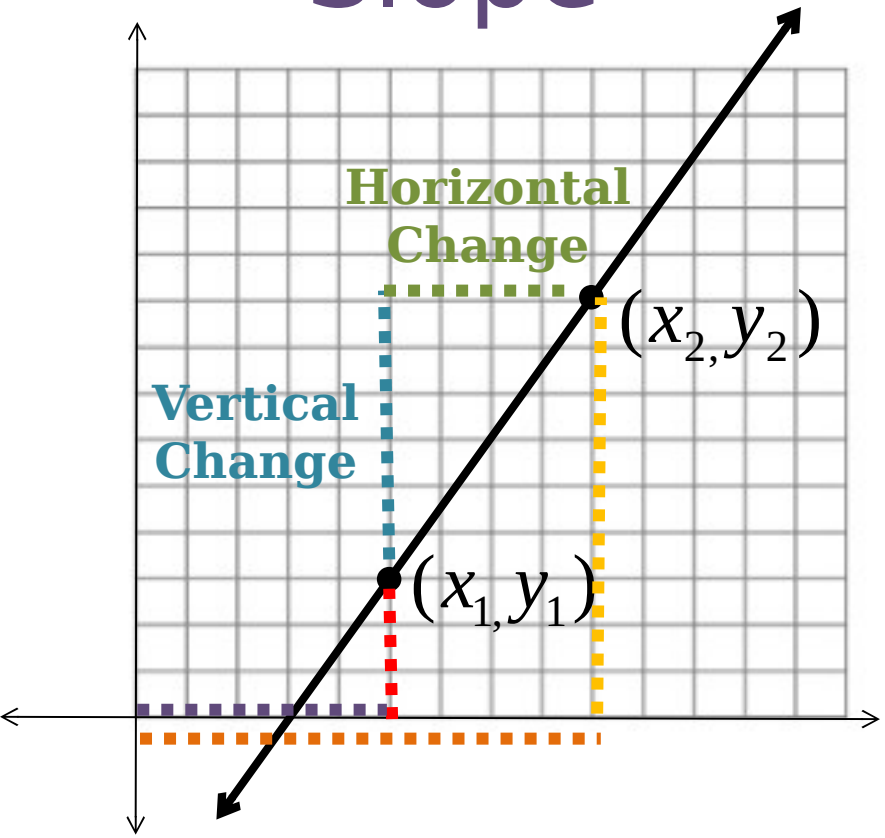
Slope



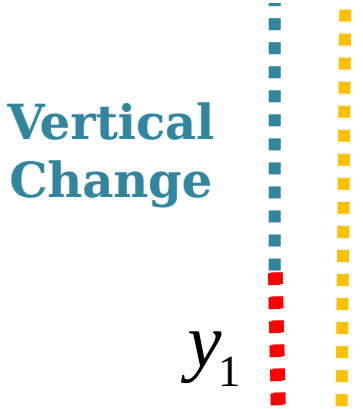
Using the slope formula

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Slope

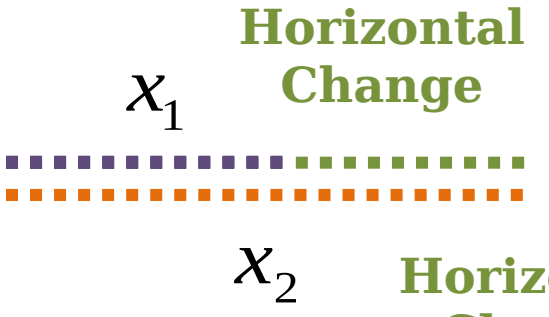


To Find Vertical Change



$$\text{Vertical Change} = y_2 - y_1$$

To Find Horizontal Change



$$\text{Horizontal Change} = x_2 - x_1$$

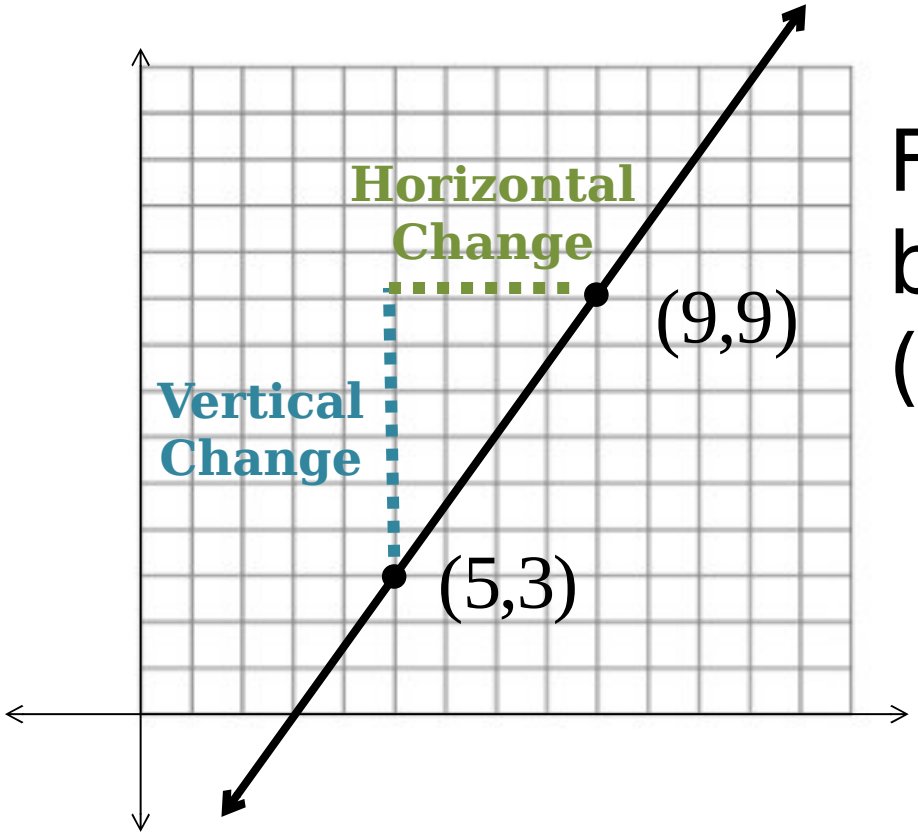
What are the lengths of the yellow and red lines?

What are the lengths of the orange and purple lines?

$$\text{Slope} = \frac{\text{Vertical Change}}{\text{Horizontal Change}} = \frac{(y_2 - y_1)}{(x_2 - x_1)}$$

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Example



Find the slope between (5, 3) and (9,9)

$$\text{Slope} = \frac{\text{Vertical Change}}{\text{Horizontal Change}} = \frac{(y_2 - y_1)}{(x_2 - x_1)}$$

$$= \frac{(9 - 3)}{(9 - 5)} = \frac{6}{4} = \frac{3}{2}$$

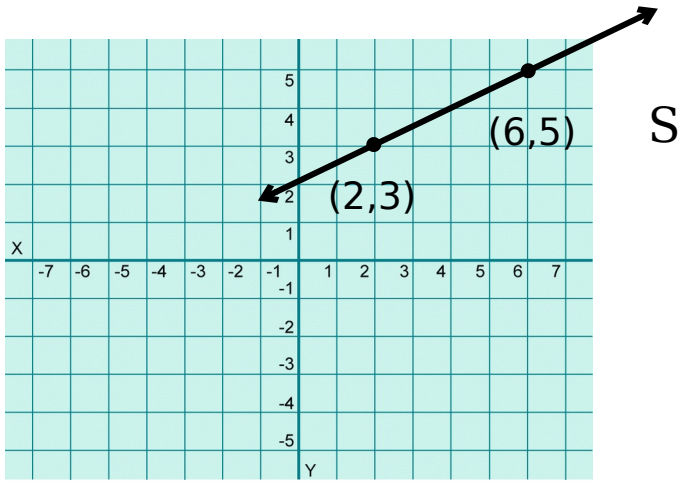
or

$$= \frac{(3 - 9)}{(5 - 9)} = \frac{-6}{-4} = \frac{3}{2}$$

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You Try It!

Find the slope between (2, 3) and (6, 5)



$$\text{Slope} = \frac{\text{Vertical Change}}{\text{Horizontal Change}} = \frac{(y_2 - y_1)}{(x_2 - x_1)}$$

$$= \frac{(5 - 3)}{(6 - 2)} = \frac{2}{4} = \frac{1}{2}$$

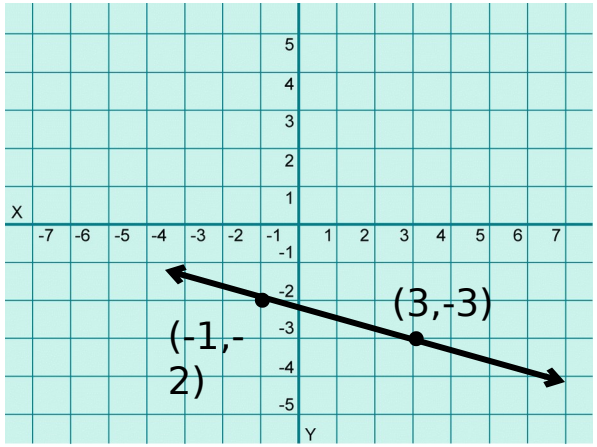
or

$$= \frac{(3 - 5)}{(2 - 6)} = \frac{-2}{-4} = \frac{1}{2}$$

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You Try It!

Find the slope between (-1, -2) and (3,-3)



$$\text{Slope} = \frac{\text{Vertical Change}}{\text{Horizontal Change}} = \frac{(y_2 - y_1)}{(x_2 - x_1)}$$

$$= \frac{(-3-(-2))}{(3-(-1))} = \frac{-1}{4} = -\frac{1}{4}$$

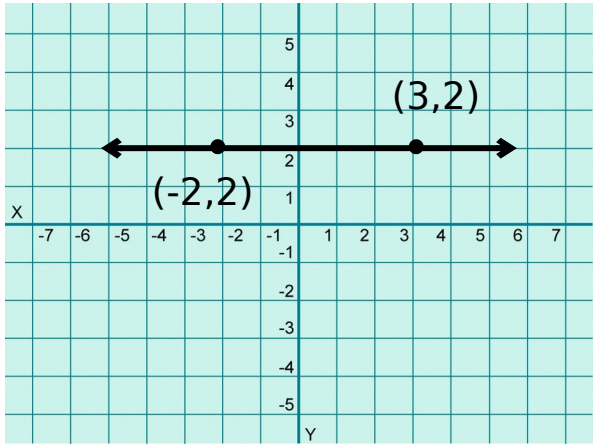
or

$$= \frac{(-2-(-3))}{(-1-3)} = \frac{1}{-4} = -\frac{1}{4}$$

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You Try It!

Find the slope between (-2, 2) and (3,2)



$$\text{Slope} = \frac{\text{Vertical Change}}{\text{Horizontal Change}} = \frac{(y_2 - y_1)}{(x_2 - x_1)}$$

$$= \frac{(2 - 2)}{(3 - (-2))} = \frac{0}{5} = 0$$

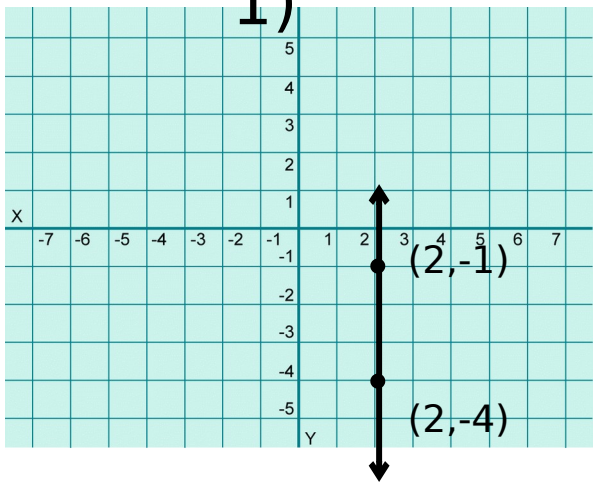
or

$$= \frac{(2 - 2)}{(-2 - 3)} = \frac{0}{-5} = 0$$

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You Try It!

1. Find the slope between (2, -4) and (2,-1)



$$\text{Slope} = \frac{\text{Vertical Change}}{\text{Horizontal Change}} = \frac{(y_2 - y_1)}{(x_2 - x_1)}$$

$$= \frac{(-1-(-4))}{(2-2)} = \frac{3}{0} = \text{undefined}$$

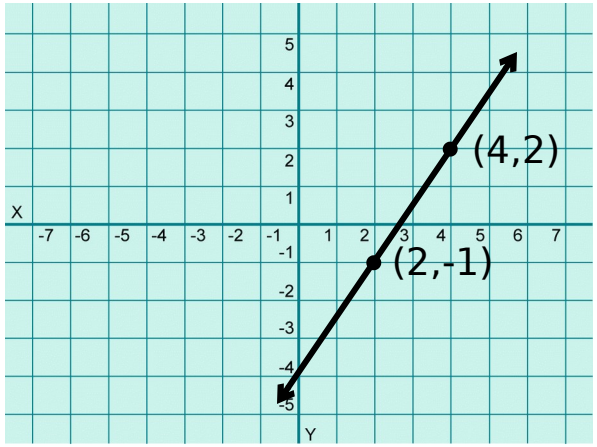
or

$$= \frac{(-4-(-1))}{(2-2)} = \frac{-3}{0} = \text{undefined}$$

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You Try It!

Find the slope between (2, -1) and (4, 2)



$$\text{Slope} = \frac{\text{Vertical Change}}{\text{Horizontal Change}} = \frac{(y_2 - y_1)}{(x_2 - x_1)}$$

$$= \frac{(2 - (-1))}{(4 - 2)} = \frac{3}{2}$$

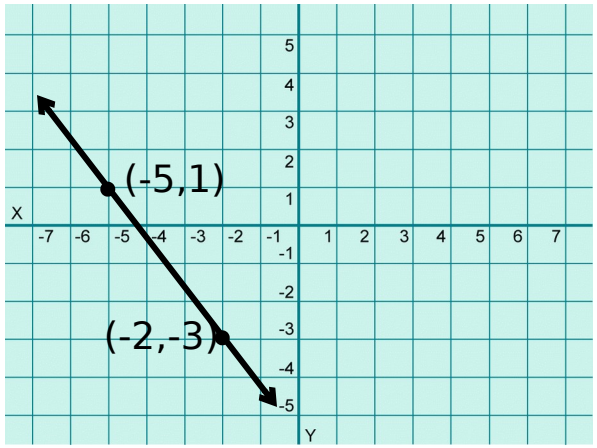
or

$$= \frac{(-1 - 2)}{(2 - 4)} = \frac{-3}{-2} = \frac{3}{2}$$

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You Try It!

Find the slope between (-5, 1) and (-2, -3)



$$\text{Slope} = \frac{\text{Vertical Change}}{\text{Horizontal Change}} = \frac{(y_2 - y_1)}{(x_2 - x_1)}$$

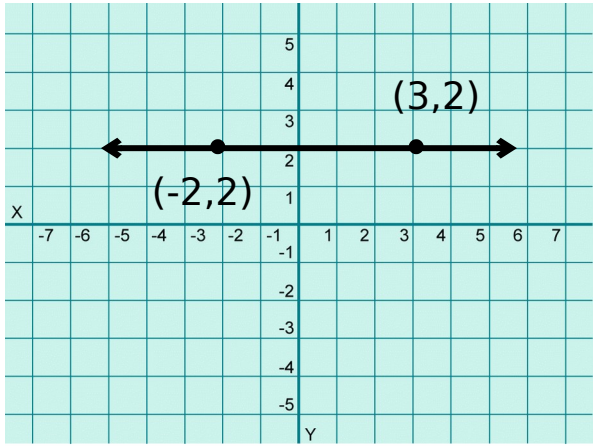
$$= \frac{(-3-1)}{(-2-(-5))} = \frac{-4}{3} = -\frac{4}{3}$$

or

$$= \frac{(1-(-3))}{(-5-(-2))} = \frac{4}{-3} = -\frac{4}{3}$$

You Try It!

Find the slope between (-2, 2) and (3,2)



$$\text{Slope} = \frac{\text{Vertical Change}}{\text{Horizontal Change}} = \frac{(y_2 - y_1)}{(x_2 - x_1)}$$

$$= \frac{(2 - 2)}{(3 - (-2))} = \frac{0}{5} = 0$$

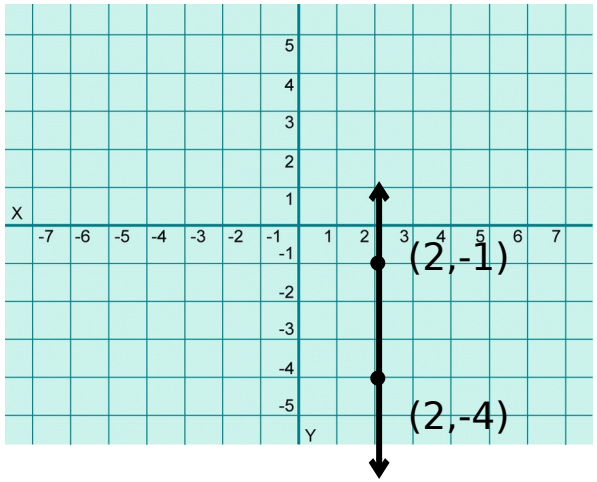
or

$$= \frac{(2 - 2)}{(-2 - 3)} = \frac{0}{-5} = 0$$

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You Try It!

Find the slope between (2, -4) and (2,-1)



$$\text{Slope} = \frac{\text{Vertical Change}}{\text{Horizontal Change}} = \frac{(y_2 - y_1)}{(x_2 - x_1)}$$

$$= \frac{(-1-(-4))}{(2-2)} = \frac{3}{0} = \textit{undefined}$$

or

$$= \frac{(-4-(-1))}{(-2-2)} = \frac{-3}{0} = \textit{undefined}$$

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Let’s play a GAME to practice!

Directions

Calculate the slope between the two given points, find it and circle it on your paper faster than your opponent!

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Find the slope between the two points.

$$m = \frac{(y_2 - y_1)}{(x_2 - x_1)}$$

1. (2,4) and (5, 9)

   5/3

2. (4, 7) and (6, 8)

   1/2

3. (6, 3) and (7, 5)

   2

4. (3, 0) and (11, 15)

   15/8

5. (6, 10) and (15, 15)

   5/9

6. (1, 5) and (5, 8)

   3/4

7. (2, 1) and (1, 3)

   -2

8. (6, 7) and (9, 3)

   -4/3

9. (1, 3) and (4, -7)

   -10/3

10. (9, 3) and (6, 3)

    0

11. (4, 8) and (4, 16)

    -8/0 =undefined

12. (1, -3) and (4, -2)

    1/3

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Directions

Calculate the slope between the two given points, find it and circle it faster than your opponent!



1/2

5/8

undefined

-2

15/8

3

2

0

-11/10

5/3

-4/3

-4

17/5

-10/3

1/3

-8/11

4/11

1

5/9

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